Physics-Informed Neural Network for 2-D Linear Elasticity

Stress Concentration in a Rectangular Plate with a Circular Hole

JAX + Flax + Optax
FourierMLP (4x128, tanh, n_fourier=8, n_polar=6)
Domain 10,000x1,000 mm, hole R=200 mm, plane_stress
E=1,000 MPa, nu=0.30, sigma0=1 MPa
460,000 epochs, LR 1e-03->1e-05

1. Problem Formulation

1.1 Geometry and Loading

The computational domain is a rectangular plate of size L x H = 10,000 mm x 1,000 mm with a central circular hole of radius R = 200 mm. The material is linear elastic with E = 1,000 MPa and nu = 0.30, under plane_stress assumptions. A uniaxial traction sigma_0 = 1 MPa is prescribed on the right edge.

1.2 Boundary Conditions

The model applies mixed displacement and traction conditions. The left edge is clamped and the hole and outer horizontal edges are traction-free.

Left (x = 0): u = 0, v = 0 [hard-enforced via network ansatz]
Right (x = L): sigma_xx = 1 MPa, tau_xy = 0
Top (y = H): sigma_yy = 0, tau_xy = 0
Bottom (y = 0): sigma_yy = 0, tau_xy = 0
Hole surface: sigma . n = 0 [soft penalty]

In addition, a midline symmetry condition v(x, H/2) = 0 is imposed as a soft penalty to suppress antisymmetric vertical modes.

1.3 Governing Equations and Constitutive Law

With zero body force, static equilibrium is enforced in strong form.

d(sigma_xx)/dx + d(tau_xy)/dy = 0
d(tau_xy)/dx + d(sigma_yy)/dy = 0

The stress-strain relation for plane_stress is written as follows.

sigma_xx = E/(1-nu^2) * (eps_xx + nu * eps_yy)
sigma_yy = E/(1-nu^2) * (eps_yy + nu * eps_xx)
tau_xy = E/(2*(1+nu)) * gamma_xy

Here eps_xx = du/dx, eps_yy = dv/dy, and gamma_xy = du/dy + dv/dx.

1.4 Kirsch Benchmark

The Kirsch infinite-plate solution predicts stress concentration factor SCF = 3 at the hole equator. For the present loading this gives sigma_xx = 3 * sigma_0 = 3 MPa at (x_c, y_c +/- R).

Because the numerical problem uses a finite domain and a clamped left edge, the exact peak stress is expected to deviate from this analytical benchmark.

2. PINN Methodology

A Physics-Informed Neural Network approximates u(x,y) and v(x,y) by a neural network trained to minimize a composite residual loss. No labeled solution data is required.

2.1 Non-dimensionalization

The model uses normalized spatial, displacement, and stress variables.

xi = x / L, eta = y / L (inputs in [0,1] x [0, 0.1])
u_hat = u / u_ref, u_ref = sigma_0 * L / E = 10 mm
sigma_tilde = sigma / sigma_0

2.2 Hard BC Ansatz

To satisfy the clamped left boundary exactly, the displacement fields are parameterized with a multiplicative xi factor.

u_hat(xi, eta) = xi * psi_u(xi, eta)
v_hat(xi, eta) = xi * psi_v(xi, eta)

2.3 Loss Function

The weighted training objective combines PDE residual and boundary terms.

L = 10*L_pde + 60*L_right + 50*L_tb + 12*L_hole + 30*L_mid

Each term is a mean-squared residual on randomly sampled collocation points for that region. Points are resampled every 2,000 epochs.

2.4 Optimization

Optimization is performed with Adam using a cosine-decay learning-rate schedule from 1e-03 to 1e-05, with warm-up over 1,000 steps, for a total of 460,000 epochs.

2.5 Network Architecture

A Fourier feature embedding is prepended to a fully-connected MLP to overcome spectral bias.

2.5.1 Input feature map (total dimension 48)

The input embedding concatenates three feature groups. The first group is the raw coordinates (xi, eta) with dimension 2. The second group contains 32 Fourier features for k = 1,...,8. The third group contains 14 hole-centered polar features for k = 1,...,6.

2.5.2 MLP backbone

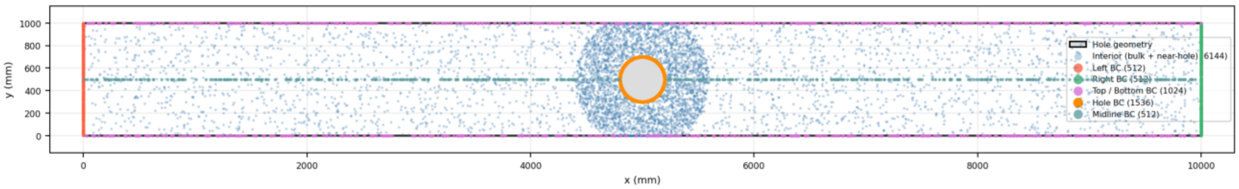

Figure 1. Training collocation points, including near-hole enrichment and boundary sampling.

3. Results

The trained model recovers the expected qualitative structure of the elasticity solution. The displacement fields remain smooth away from the hole, while the stress plots show clear concentration and redistribution around the circular boundary. The loss history provides a complementary view of optimization behavior, including stochastic variability induced by random collocation resampling.

3.1 Training History

The total loss decreases substantially over training, while individual loss components settle to different magnitudes depending on weighting and physical difficulty.

3.2 Full-Field Predictions

Full-domain displacement and stress plots show the far-field loading response together with the perturbation induced by the hole.

3.3 Near-Hole and Deformed Views

Near-hole stress plots isolate the region where the strongest gradients occur. Deformed plots provide a qualitative visualization of how predicted fields distort around the cavity.

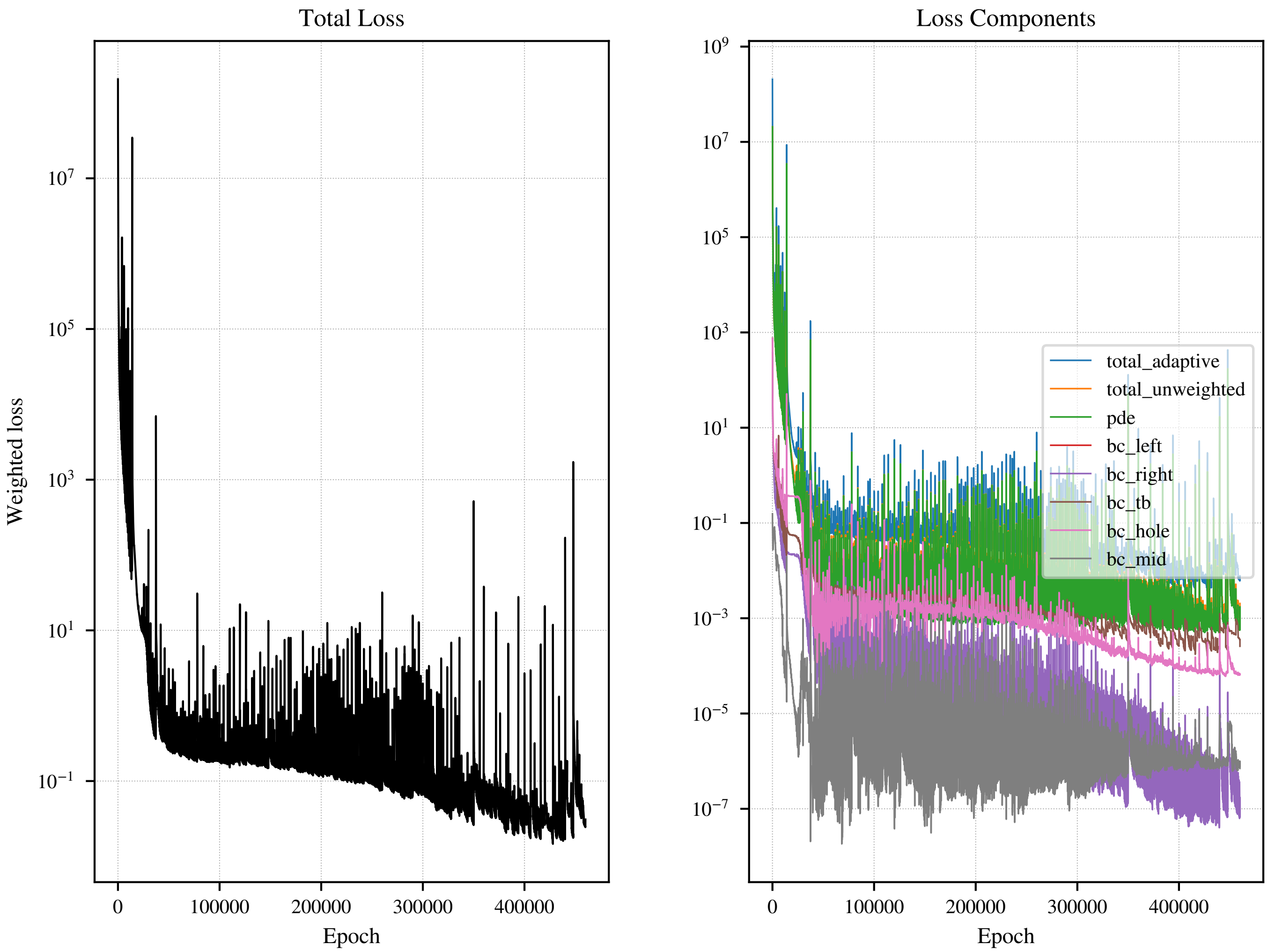

Figure 2. Training loss history on a log scale. Left: total weighted loss. Right: individual PDE and boundary loss components.

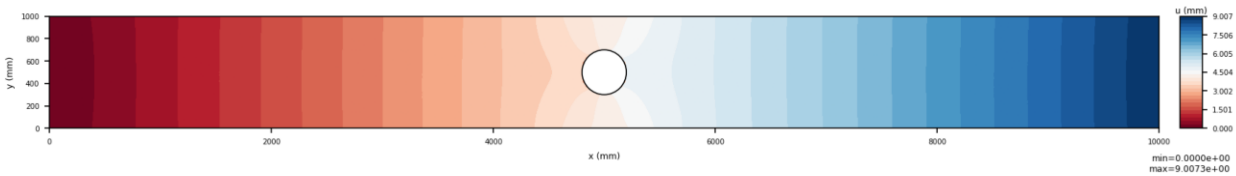

Figure 3. undeformed u.

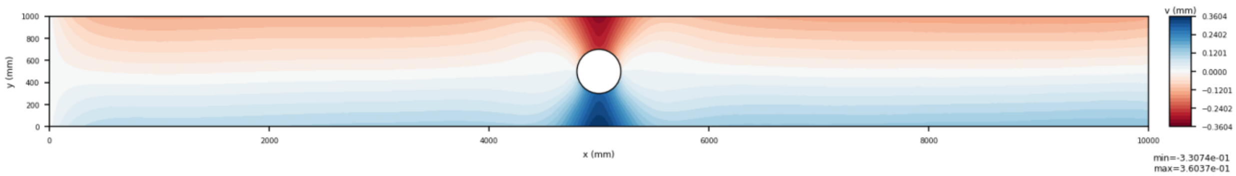

Figure 4. undeformed v.

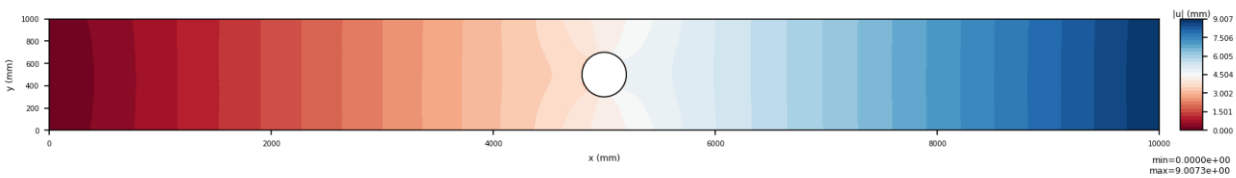

Figure 5. undeformed umag.

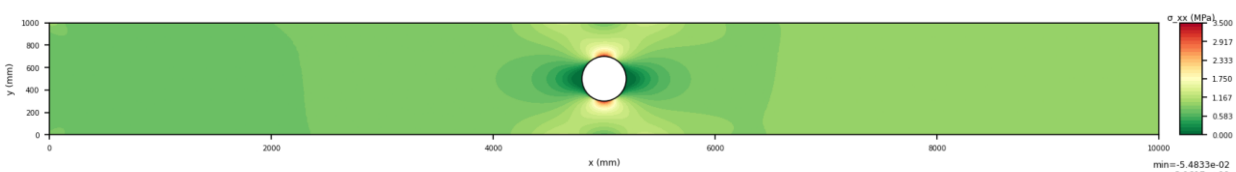

Figure 6. undeformed σ_xx.

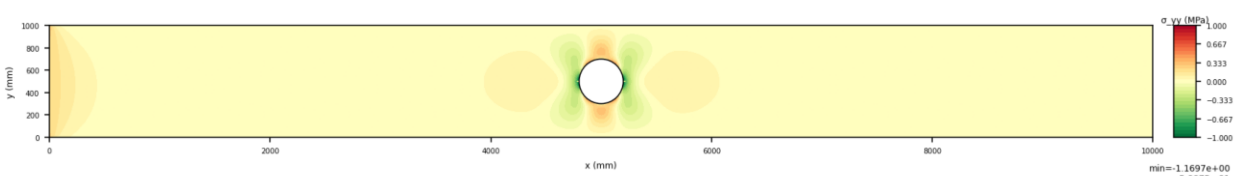

Figure 7. undeformed σ_yy.

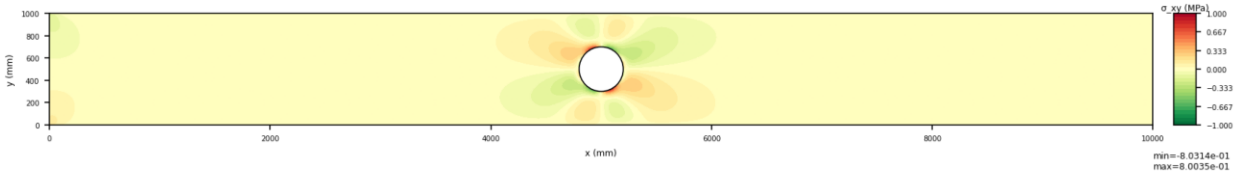

Figure 8. undeformed σ_xy.

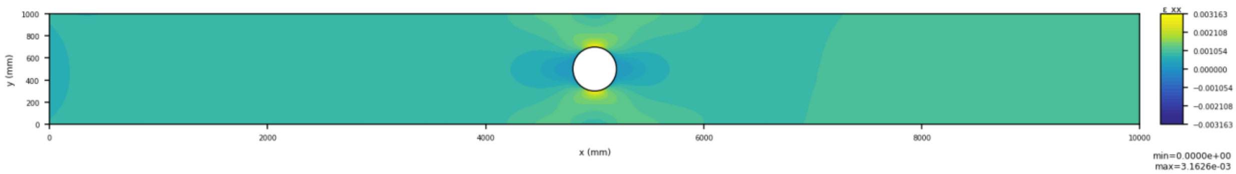

Figure 9. undeformed ε_xx.

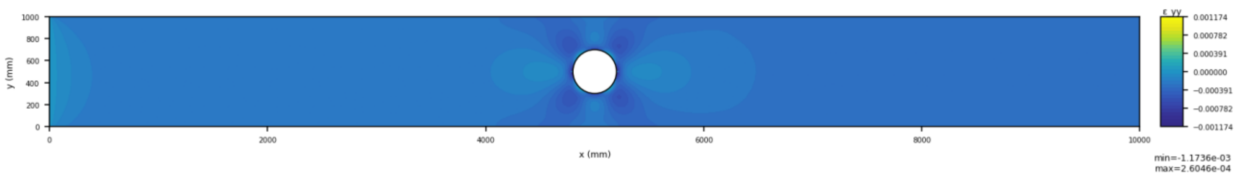

Figure 10. undeformed ε_yy.

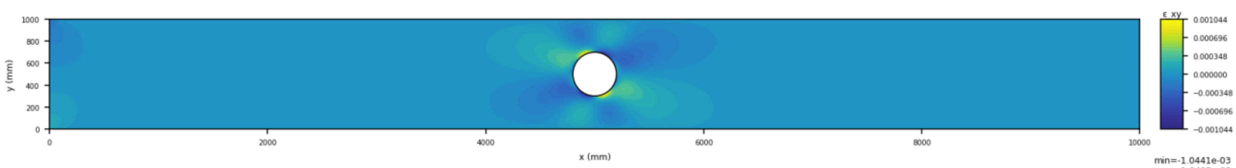

Figure 11. undeformed ε_xy.

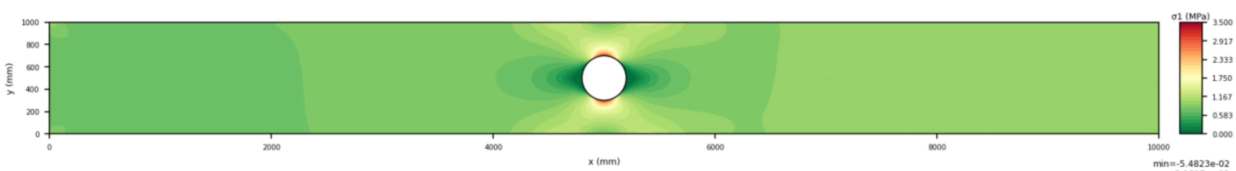

Figure 12. undeformed σ1.

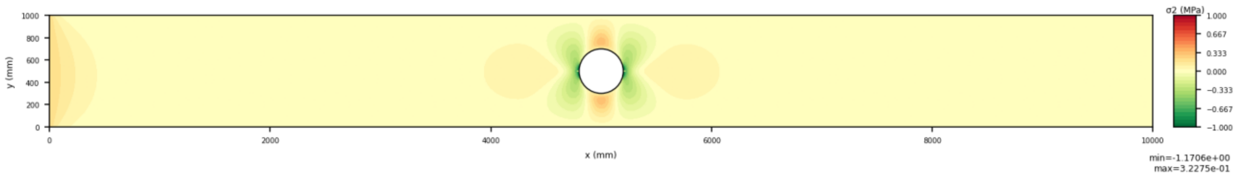

Figure 13. undeformed σ2.

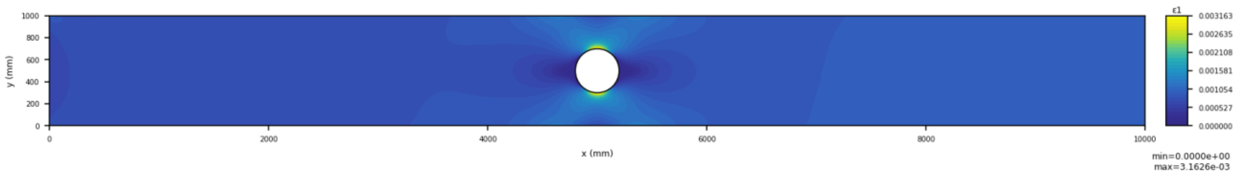

Figure 14. undeformed ε1.

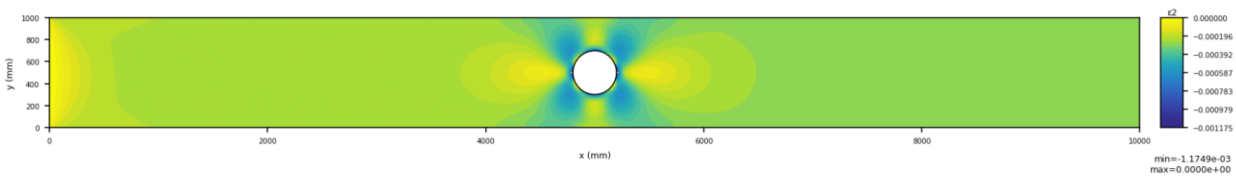

Figure 15. undeformed ε2.

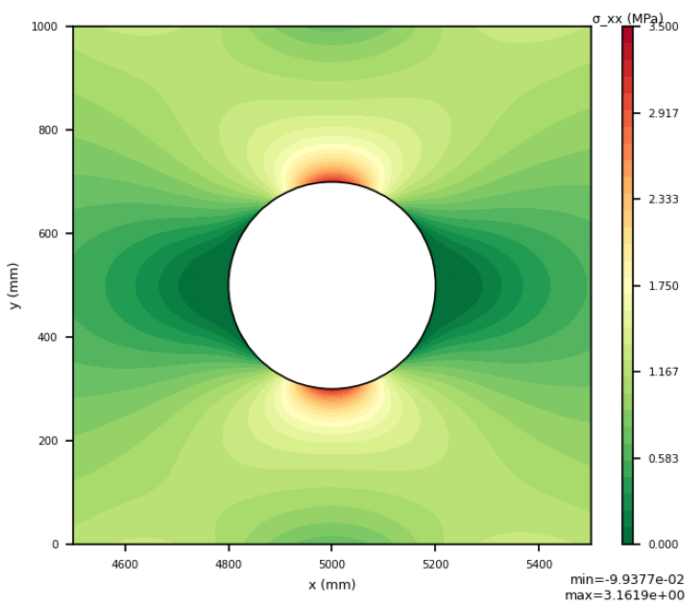

Figure 16. undeformed zoom σ_xx.

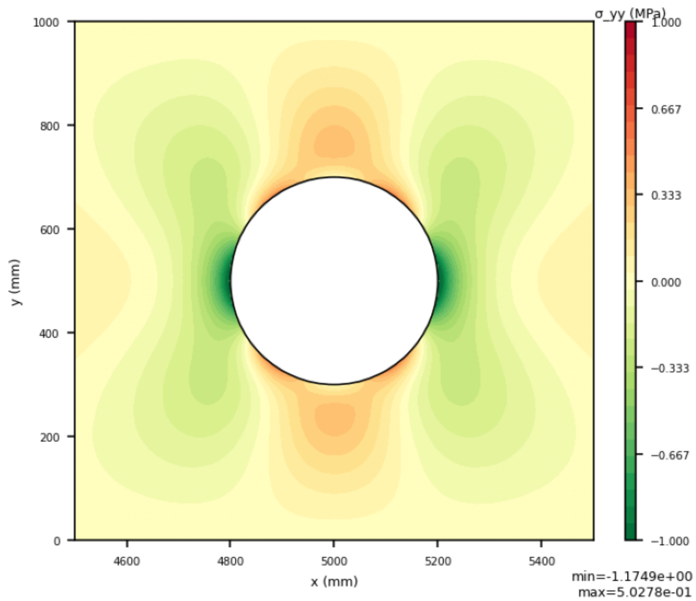

Figure 17. undeformed zoom σ_yy.

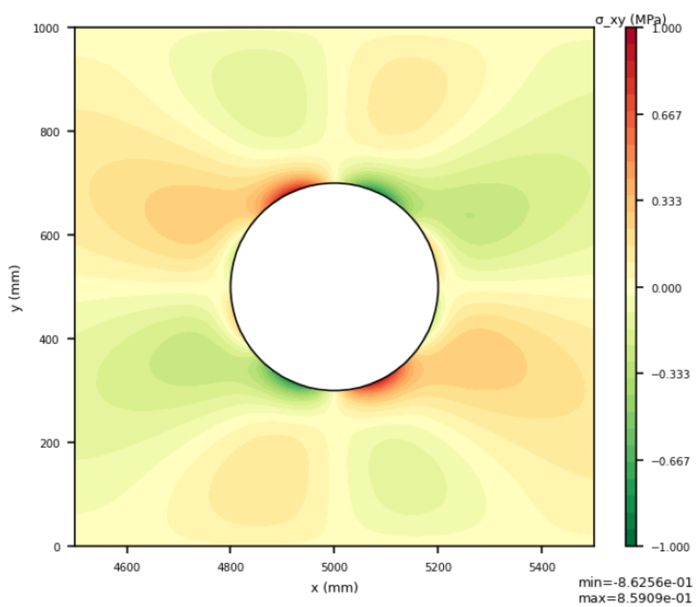

Figure 18. undeformed zoom σ_xy.

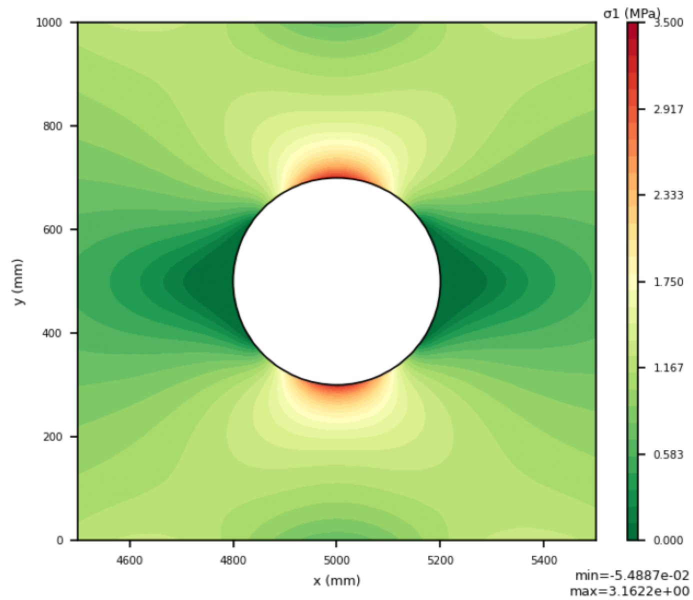

Figure 19. undeformed zoom σ1.

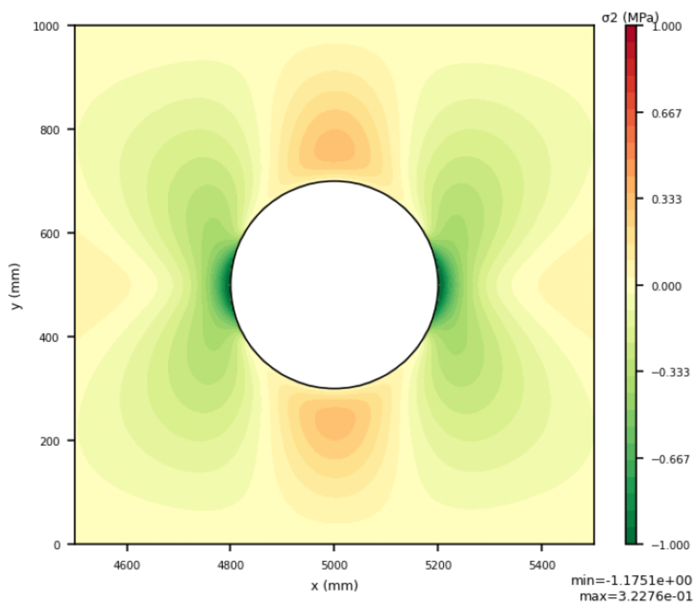

Figure 20. undeformed zoom σ2.

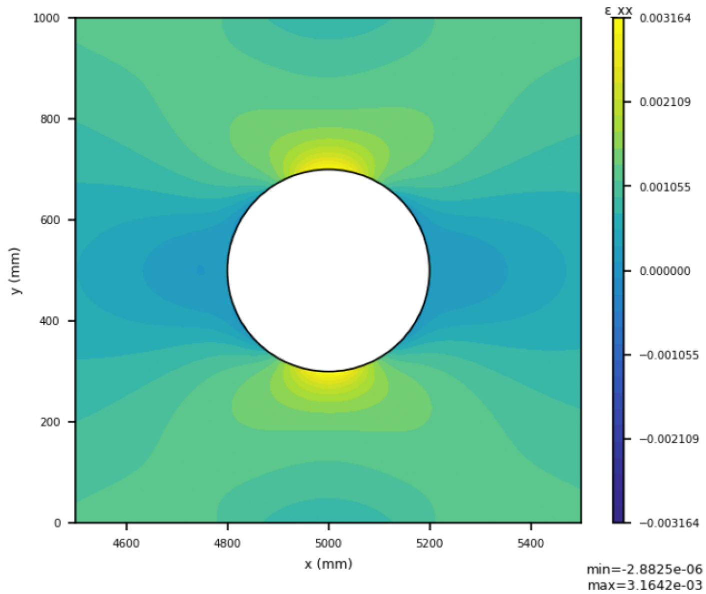

Figure 21. undeformed zoom ε_xx.

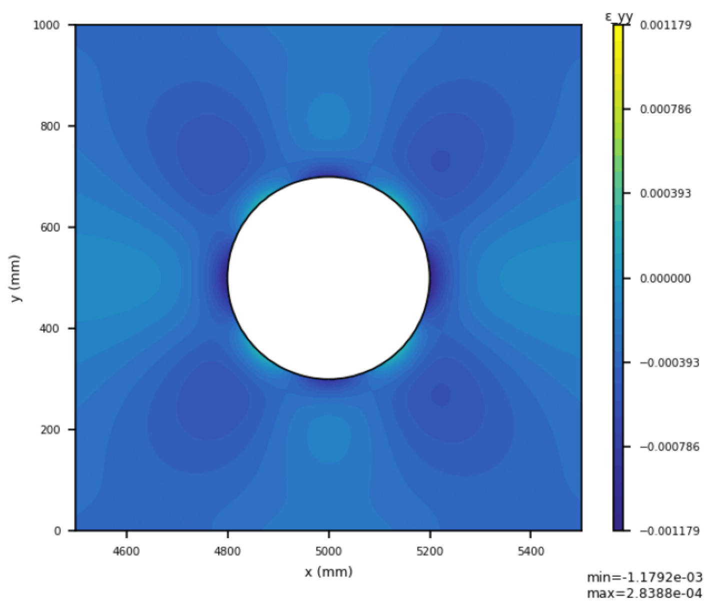

Figure 22. undeformed zoom ε_yy.

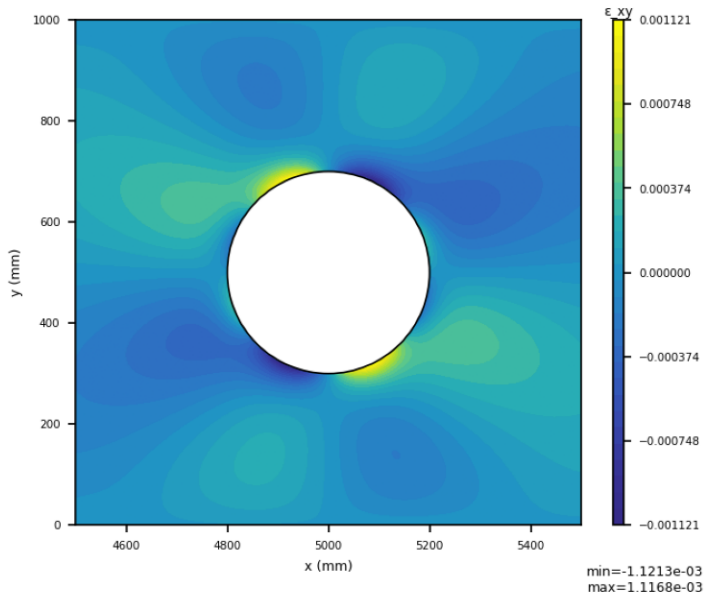

Figure 23. undeformed zoom ε_xy.

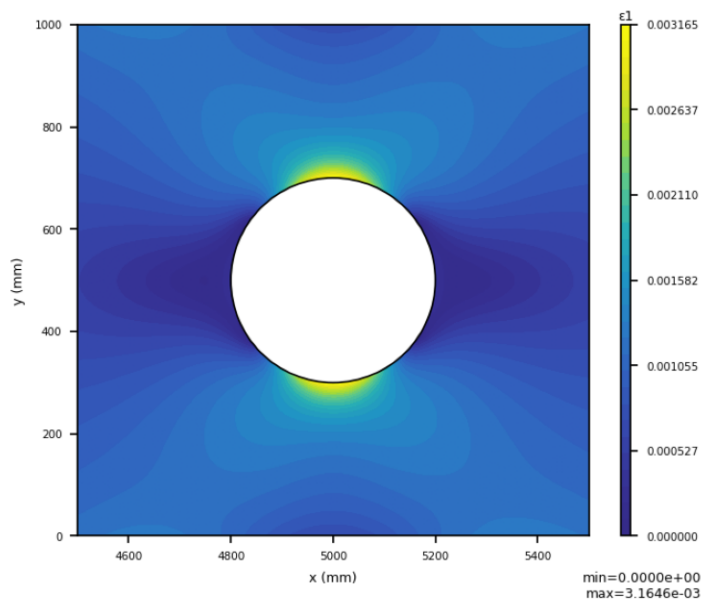

Figure 24. undeformed zoom ε1.

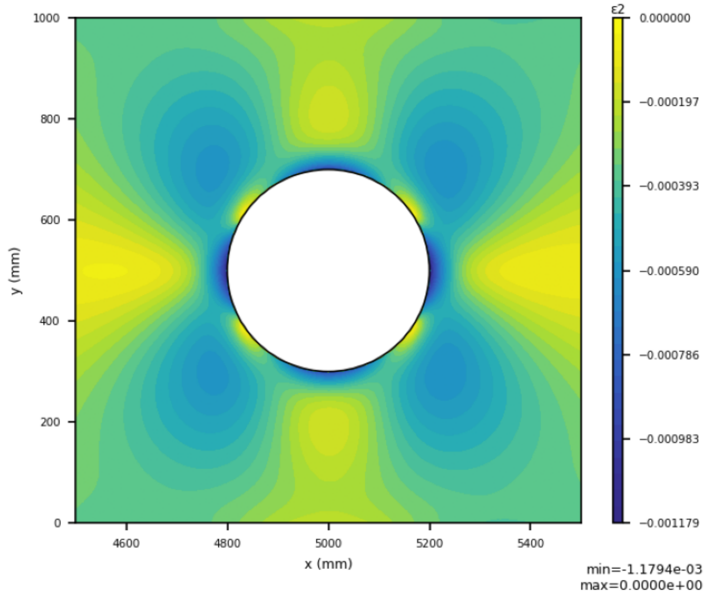

Figure 25. undeformed zoom ε2.

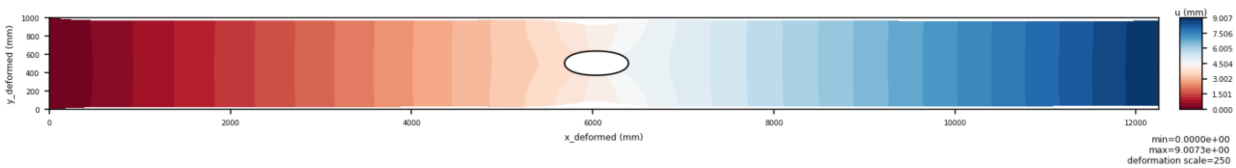

Figure 26. deformed u.

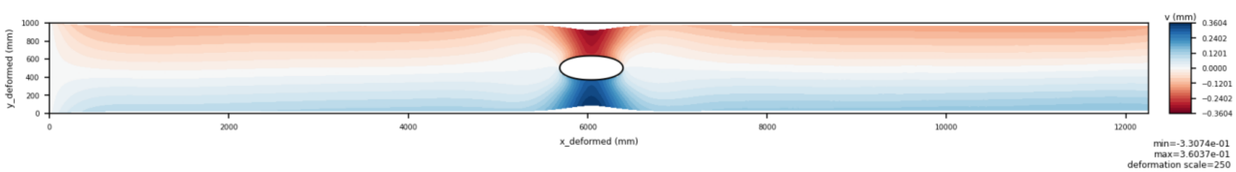

Figure 27. deformed v.

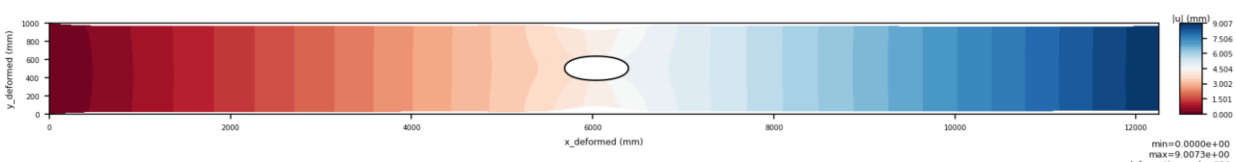

Figure 28. deformed umag.

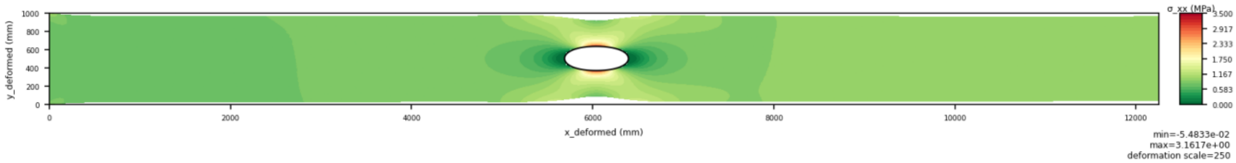

Figure 29. deformed σ_xx.

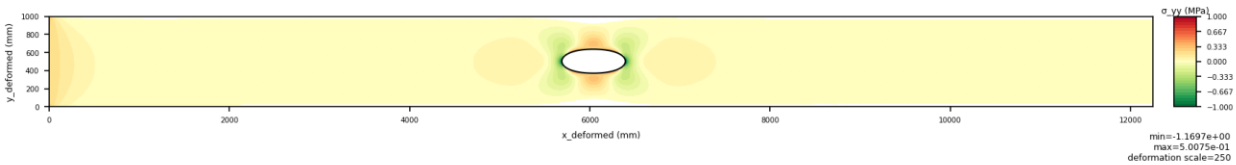

Figure 30. deformed σ_yy.

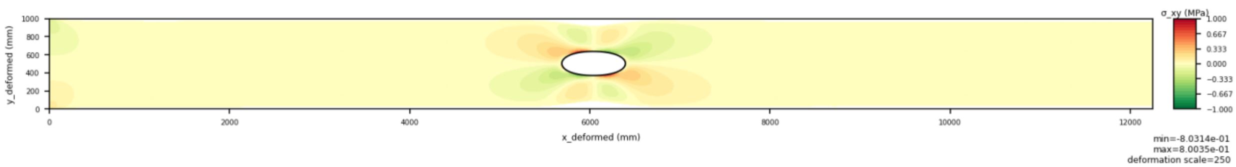

Figure 31. deformed σ_xy.

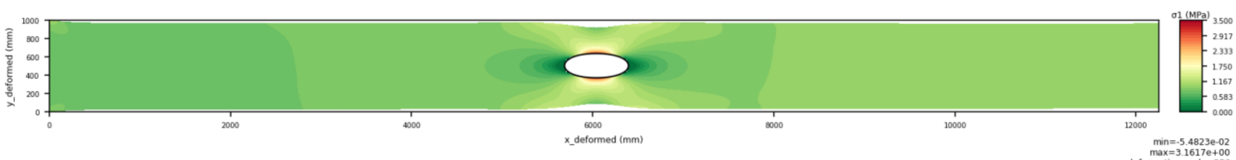

Figure 32. deformed σ1.

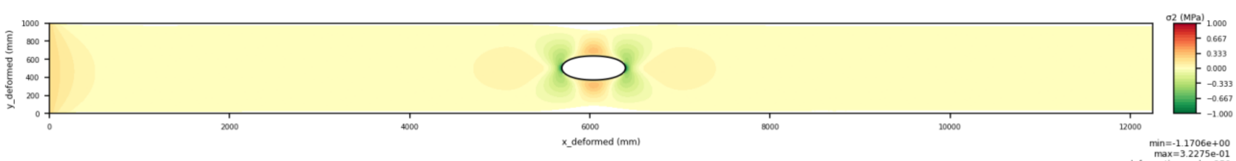

Figure 33. deformed σ2.

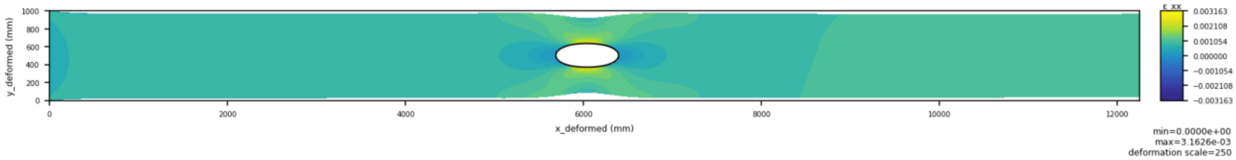

Figure 34. deformed ε_xx.

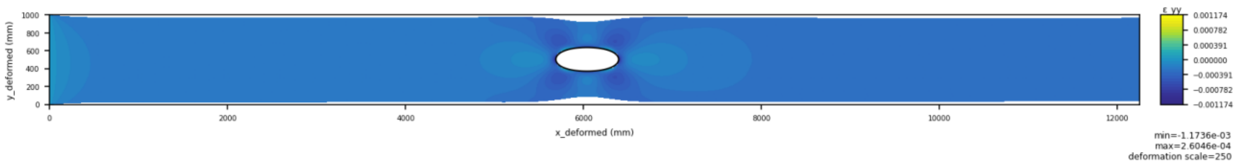

Figure 35. deformed ε_yy.

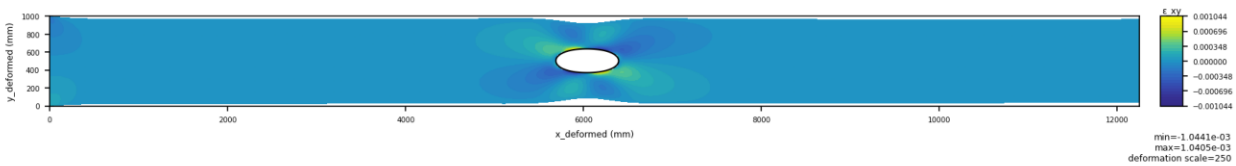

Figure 36. deformed ε_xy.

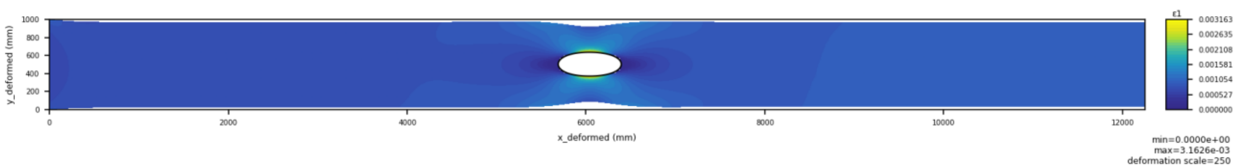

Figure 37. deformed ε1.

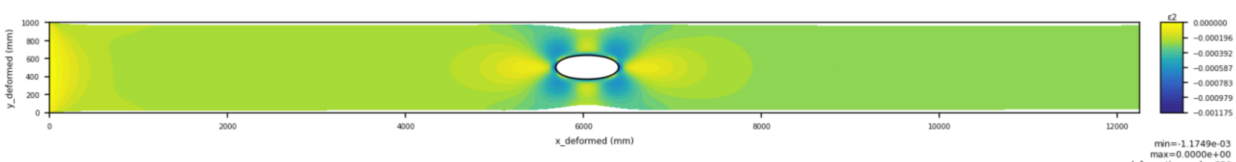

Figure 38. deformed ε2.

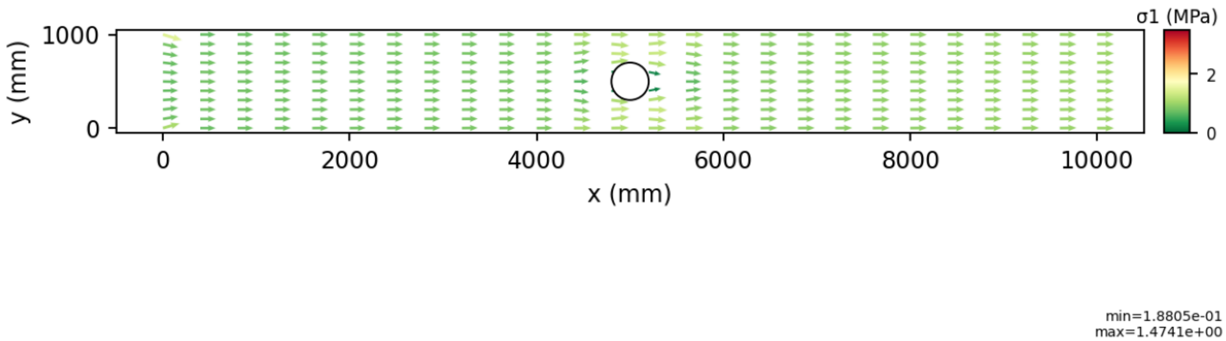

Figure 39. undeformed vector σ1.

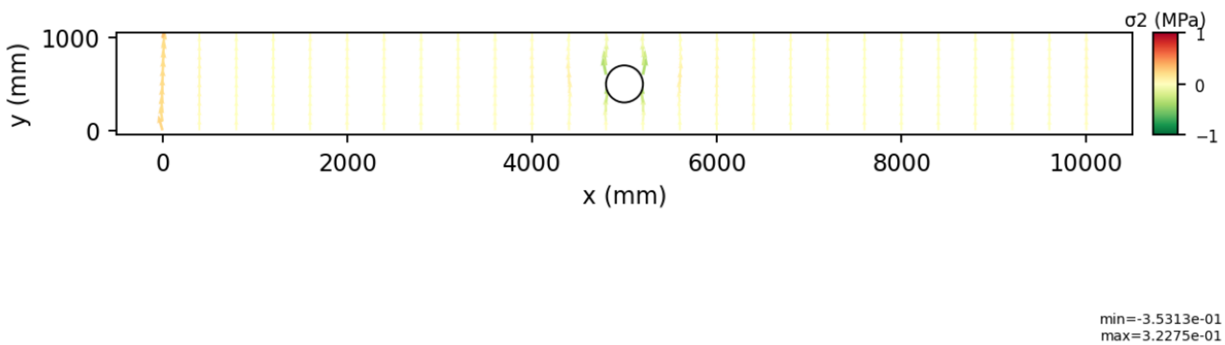

Figure 40. undeformed vector σ2.

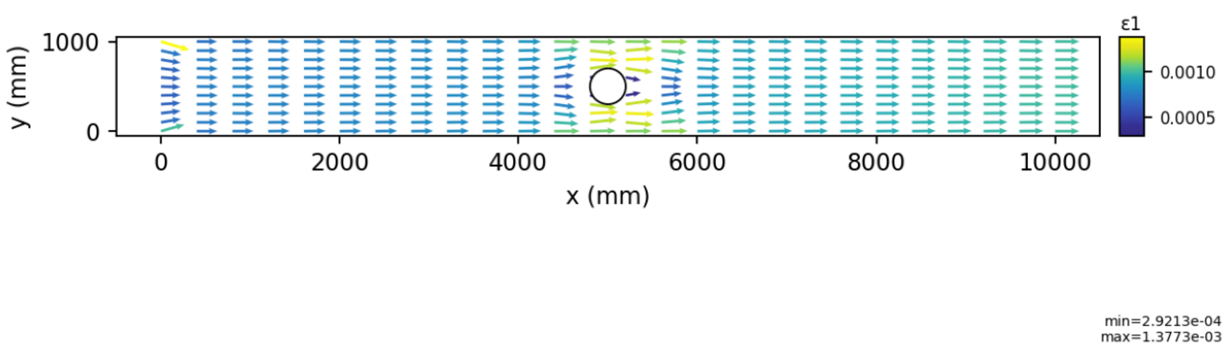

Figure 41. undeformed vector ε1.

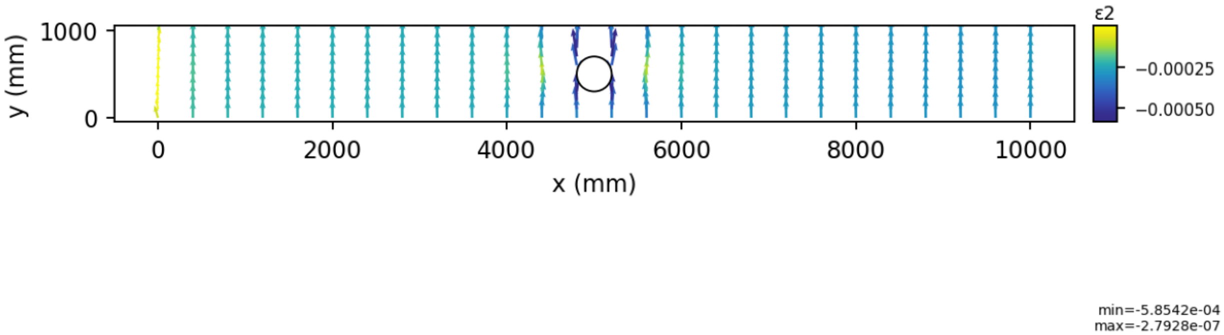

Figure 42. undeformed vector ε2.

4. Discussion and Conclusions

The FourierMLP PINN resolves stress concentration around the circular hole and reproduces the expected sigma_xx peak behavior near the hole equator. The Fourier and polar embeddings improve near-hole expressivity, and the hard boundary-condition ansatz stabilizes optimization by enforcing the left-edge clamp exactly.

With sigma_0 = 1 MPa, the Kirsch infinite-plate target is sigma_xx,peak = 3*sigma_0 = 3 MPa (SCF = 3). The present results should therefore be interpreted against that reference: agreement indicates correct concentration magnitude, while any mismatch from 3 MPa is expected to reflect finite-domain effects and the clamped boundary at $x = 0$, not only network approximation error.

At the same time, the optimization history remains noisy because stochastic collocation introduces variance in residual estimates. The near-hole stress field is qualitatively correct, but the report still lacks a systematic quantitative error study against a trusted FEM or analytical reference over the finite domain.

Overall, the present methodology is effective as a research prototype and produces physically plausible fields, but quantitative verification and robustness should be strengthened before predictive use.

5. Future Work

A natural next step is to compare PINN predictions against a finite-element reference solution on the same geometry and loading case, so displacement and stress errors can be reported quantitatively.

Another useful extension is adaptive loss reweighting, such as NTK-based balancing or residual-driven weighting, to improve optimization stability and reduce manual hyperparameter tuning.

The framework can also be extended to nonlinear material behavior, more complex geometries, and eventually three-dimensional elasticity.